

Ratio Between Fermi–Amaldi and Exact Exchange Energy Densities

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I. MAIN TEXT

The goal is to establish a relationship between the exact exchange energy density (closed-shell):

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

where

$$\gamma(\mathbf{r}, \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}') \quad (2)$$

is the one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

When comparing two integrals with different integrands $f(x)$ and $g(x)$, on the interval $[a, b]$, the following inequality holds:

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx, \quad \text{when } f(x) \geq g(x). \quad (4)$$

Both integrands in Eqs. (1) and (3) share the same denominator, $1/|\mathbf{r} - \mathbf{r}'|$, which is positive everywhere in space. Therefore, by the Cauchy–Schwarz inequality:

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r}, \mathbf{r}')|^2 \quad \text{for all } \mathbf{r}, \mathbf{r}', \quad (5)$$

we obtain the following relationship:

$$\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \geq \int \frac{|\gamma(\mathbf{r}, \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad \text{for all } \mathbf{r}. \quad (6)$$

Then, by the definitions (1) and (3):

$$\frac{e_x^{\text{exact}}(\mathbf{r})}{e_x^{\text{FA}}(\mathbf{r})} \leq \frac{N}{2} \quad \text{for all } \mathbf{r}. \quad (7)$$

Since both exchange energy densities are non-positive, their ratio is non-negative, leading to the inequality:

$$0 \leq \frac{e_x^{\text{exact}}(\mathbf{r})}{e_x^{\text{FA}}(\mathbf{r})} \leq \frac{N}{2} \quad \text{for all } \mathbf{r}. \quad (8)$$

Thus, the upper bound of the ratio between the two closed-shell exchange energy densities is given by a function of the electron number. Note that for $N = 2$, the upper bound is unity [5–7].

II. ADDITIONAL DETAILS

In the asymptotic limit, the denominator of Eqs. (1) and (3) can be approximated as $|\mathbf{r} - \mathbf{r}'| \approx |\mathbf{r}|$, revealing that the exchange energy densities share the same asymptotic behavior:

$$e_x^{\text{exact}}(\mathbf{r}) \sim e_x^{\text{FA}}(\mathbf{r}) \sim -\frac{\rho(\mathbf{r})}{2|\mathbf{r}|}, \quad (9)$$

$$\lim_{|\mathbf{r}| \rightarrow \infty} \frac{e_x^{\text{FA}}(\mathbf{r})}{e_x^{\text{exact}}(\mathbf{r})} = 1. \quad (10)$$

Conversely, in the limit of infinitesimally small interelectronic distances ($\mathbf{r}' \rightarrow \mathbf{r}$), the spatial parts of the integrands approach the same value:

$$\lim_{\mathbf{r}' \rightarrow \mathbf{r}} |\gamma(\mathbf{r}, \mathbf{r}')|^2 = \lim_{\mathbf{r}' \rightarrow \mathbf{r}} \rho(\mathbf{r})\rho(\mathbf{r}') = \rho(\mathbf{r})^2. \quad (11)$$

Because the exact exchange prefactor $1/4$ is strictly larger than the Fermi–Amaldi prefactor $1/(2N)$ for $N > 2$, the exact exchange integrand strongly dominates the Fermi–Amaldi integrand in the vicinity of the reference electron ($\mathbf{r}' \approx \mathbf{r}$). This observation, however, does not imply this type of dominance of the integrated quantity, i.e., the exact exchange energy density.

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